

# Customer Churn Prediction & Retention Analytics

Python + Machine Learning + Power BI

## Executive Summary

This project analyses 7,043 telecom customers to understand churn behaviour, predict churn risk and support customer-retention decisions. The observed churn rate is 26.54% (1,869 churned customers). The machine-learning workflow compared four classifiers and selected **Gradient Boosting** by held-out ROC-AUC (0.845). The final scoring output identifies 905 high-risk customers representing \$72,337.30 in monthly charges.

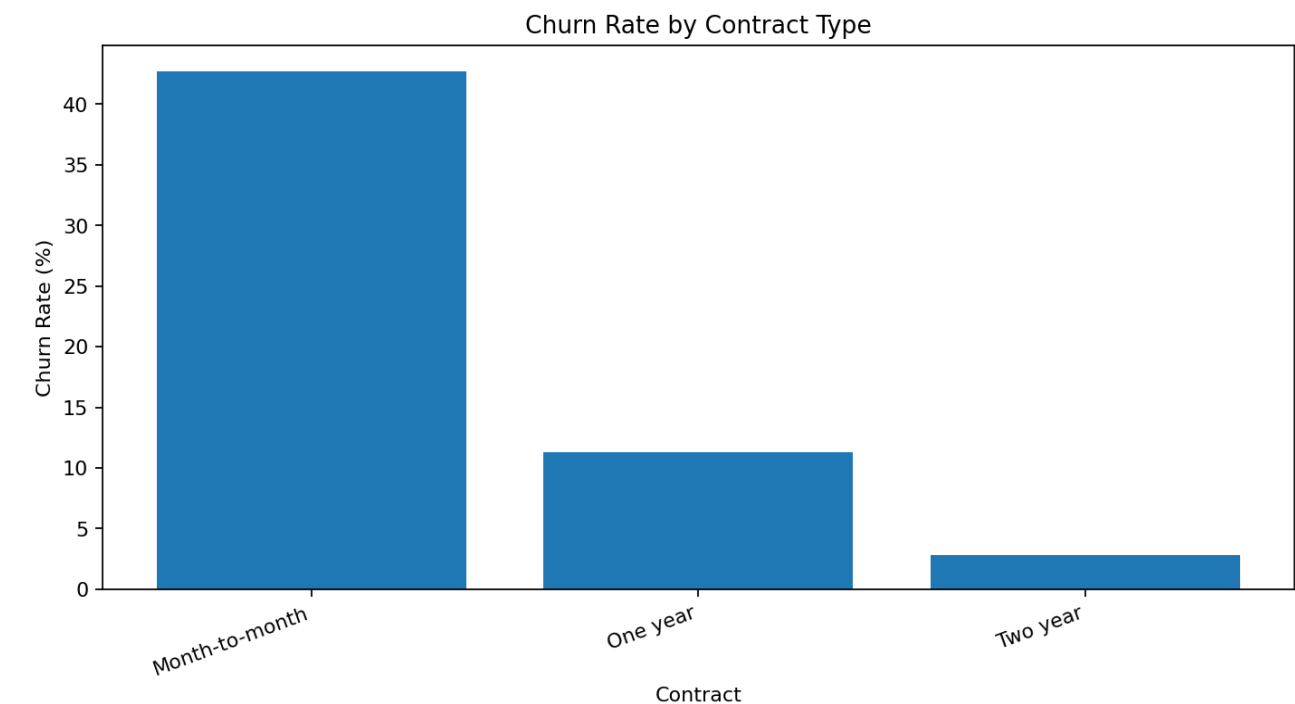
| Metric                  | Result      |
|-------------------------|-------------|
| Total customers         | 7,043       |
| Churned customers       | 1,869       |
| Retained customers      | 5,174       |
| Observed churn rate     | 26.54%      |
| High-risk customers     | 905         |
| Monthly charges at risk | \$72,337.30 |

## Data Preparation

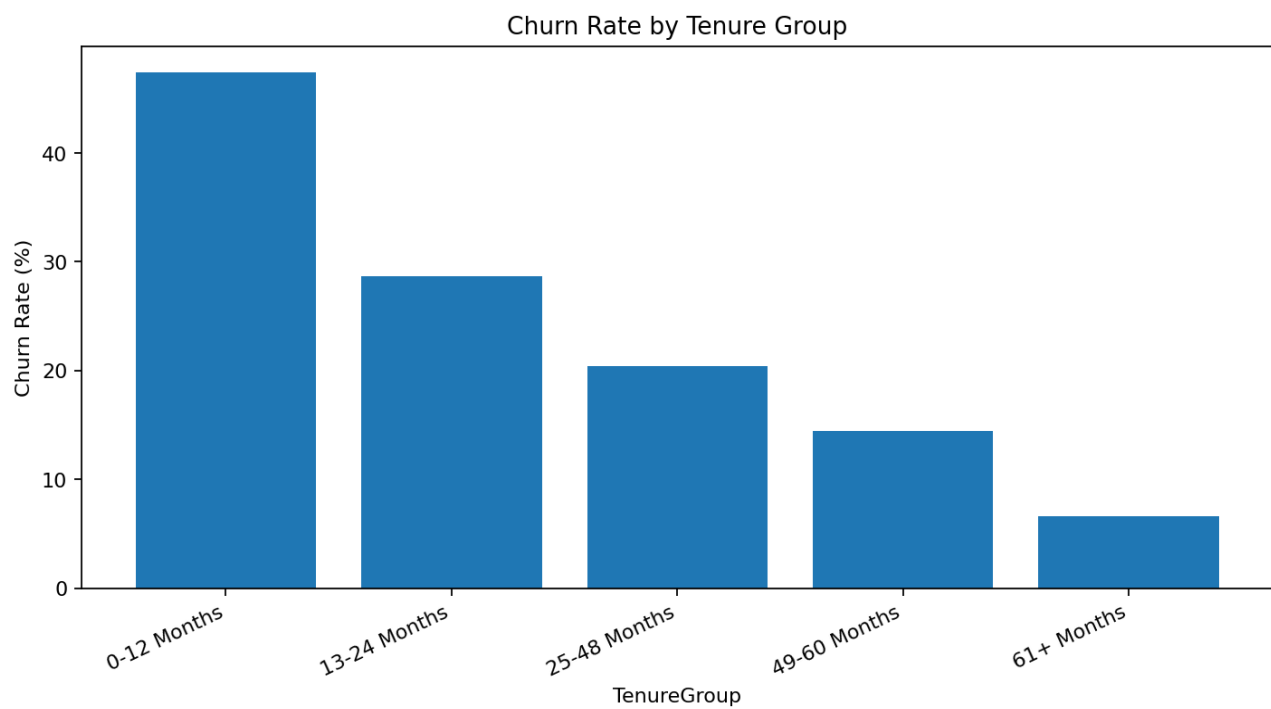
The workflow converts TotalCharges to numeric, resolves blank values, removes exact duplicates, validates customer identifiers and the churn target, and creates a binary ChurnFlag. Business-oriented features include TenureGroup, MonthlyChargeBand, ServiceCount, ContractRisk and EstimatedCustomerValue.

## Observed Churn Patterns

The strongest observed churn patterns are concentrated among month-to-month customers, electronic-check customers, fibre-optic internet customers and customers in their first 12 months. Lack of Tech Support and Online Security is also associated with substantially higher churn in this dataset.



Churn rate by contract type

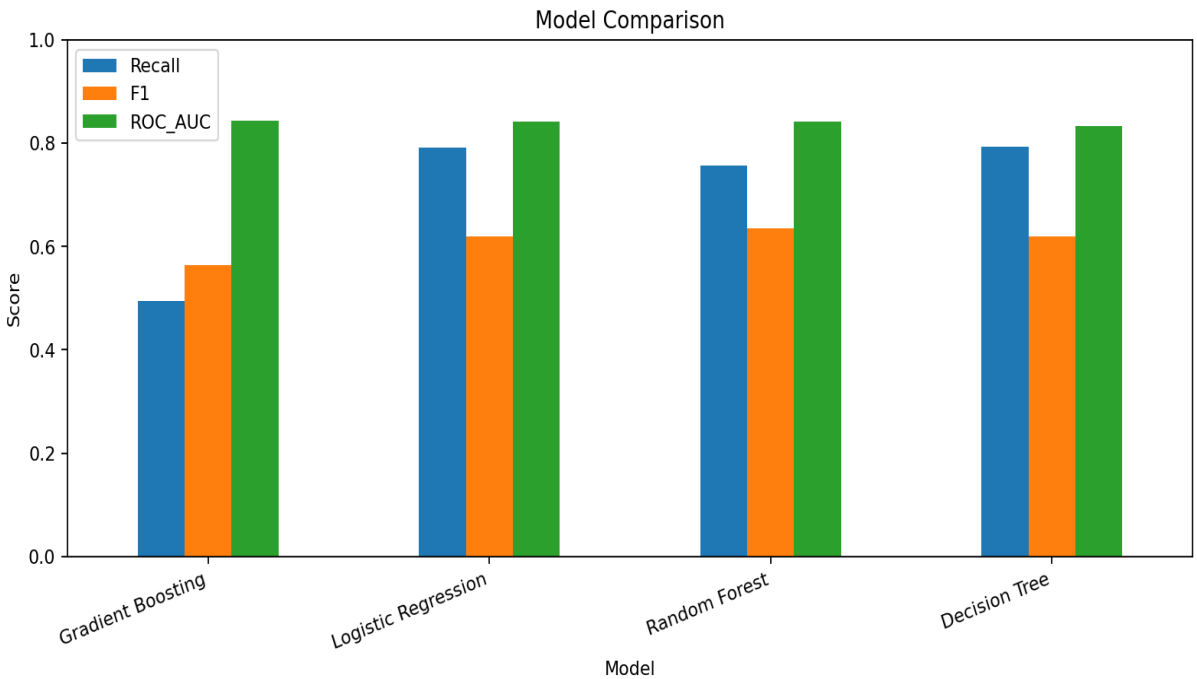


Churn rate by tenure group

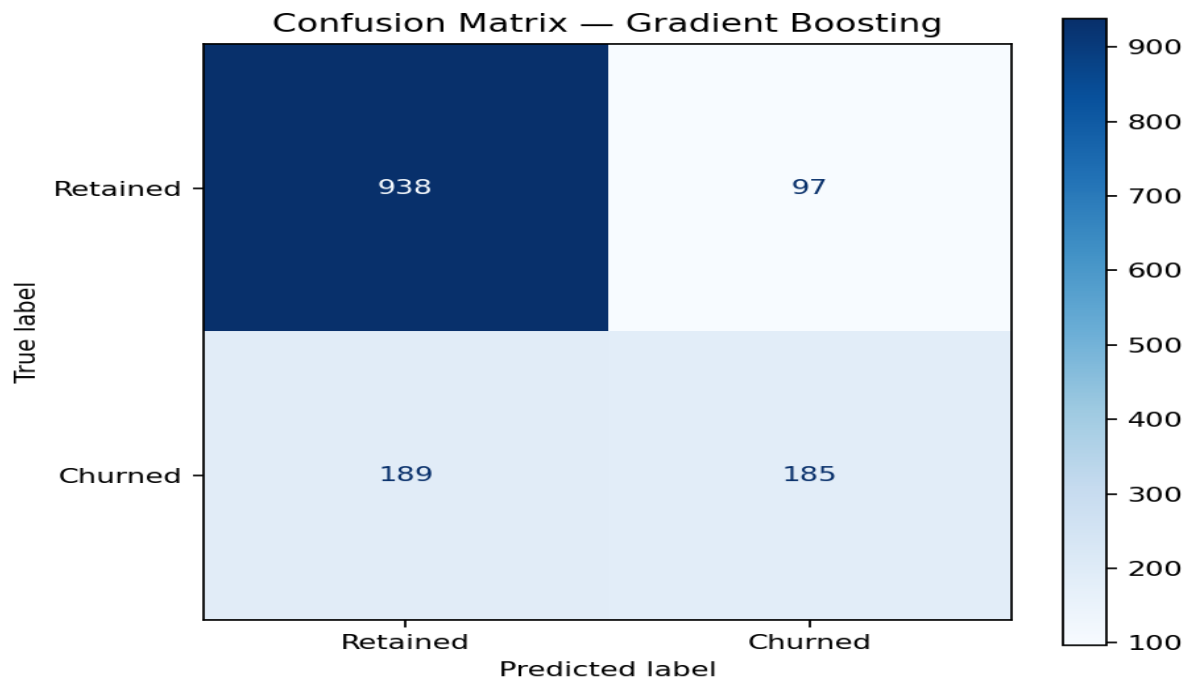
# Machine Learning Model Comparison

Four classifiers were evaluated on a stratified held-out test set using Accuracy, Precision, Recall, F1 and ROC-AUC. Gradient Boosting achieved the highest ROC-AUC and was therefore selected by the notebook. However, Logistic Regression and Random Forest produced substantially higher churn recall, which is an important consideration when the business priority is to avoid missing likely churners.

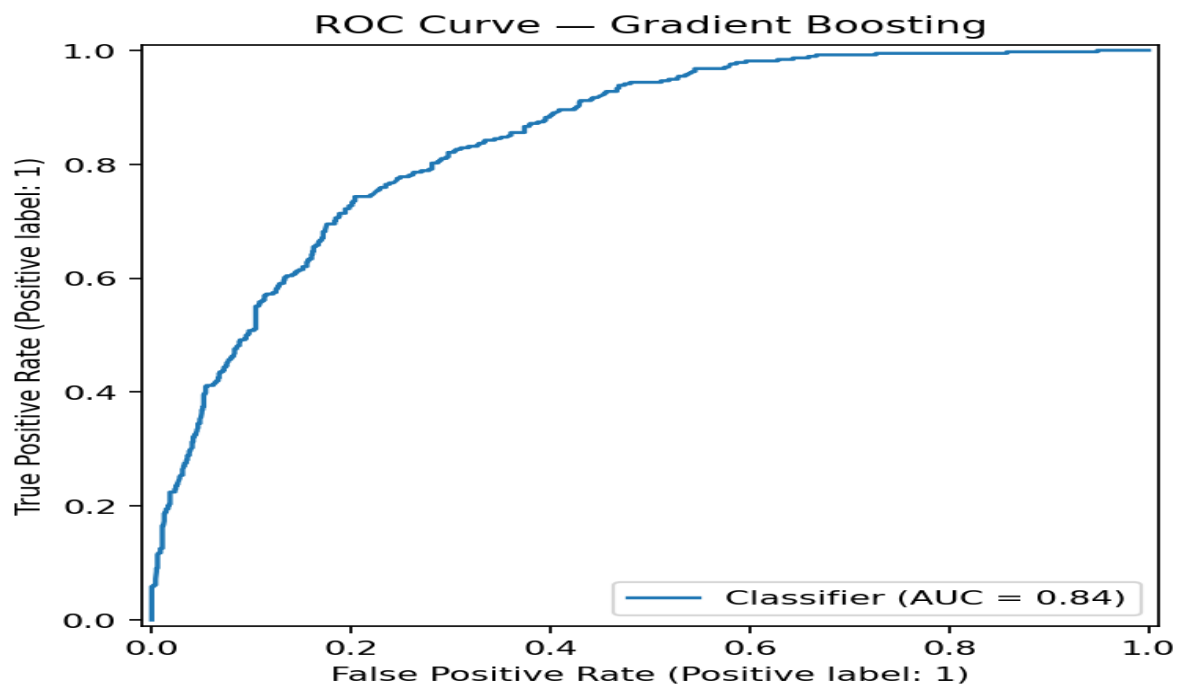
| Model               | Accuracy | Precision | Recall | F1    | ROC-AUC |
|---------------------|----------|-----------|--------|-------|---------|
| Gradient Boosting   | 0.797    | 0.656     | 0.495  | 0.564 | 0.845   |
| Logistic Regression | 0.742    | 0.509     | 0.791  | 0.620 | 0.842   |
| Random Forest       | 0.770    | 0.548     | 0.757  | 0.636 | 0.842   |
| Decision Tree       | 0.741    | 0.508     | 0.794  | 0.619 | 0.834   |



Model comparison across recall, F1 and ROC-AUC



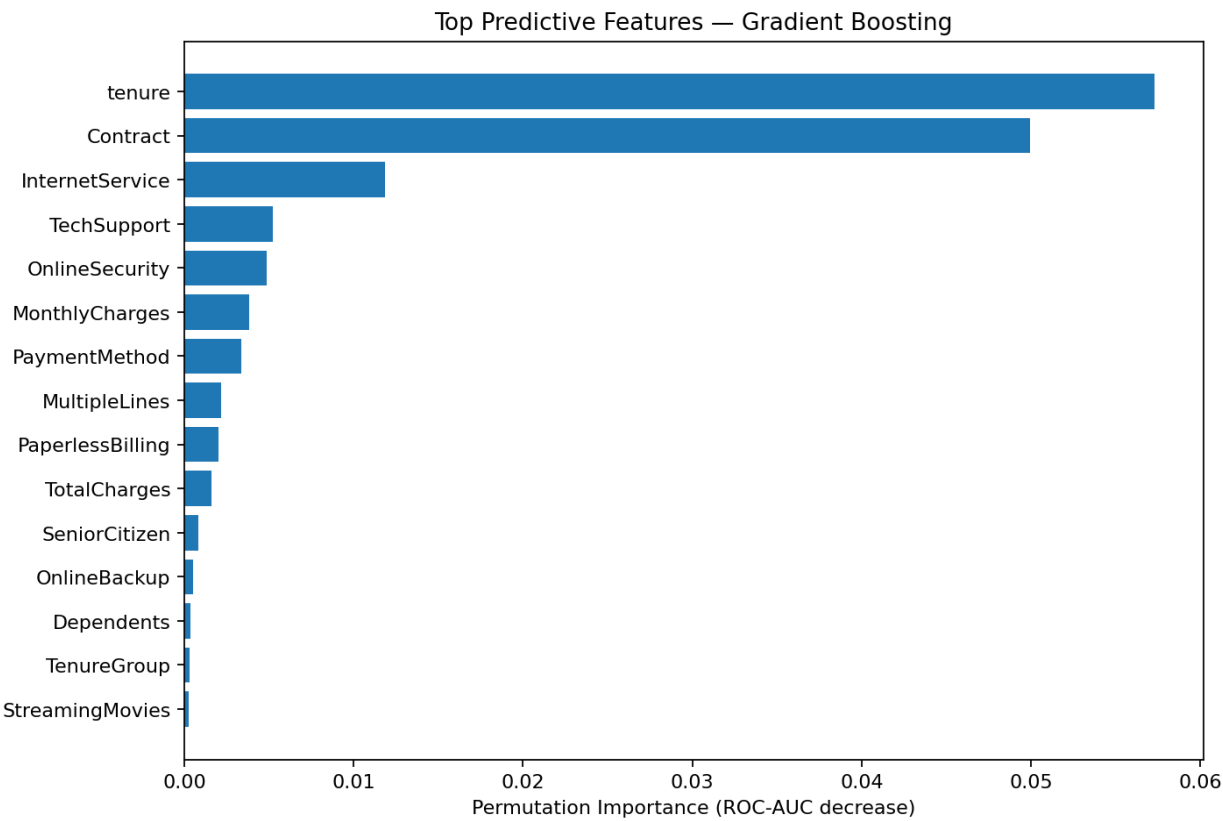
Confusion matrix for Gradient Boosting



ROC curve for Gradient Boosting

## Feature Importance

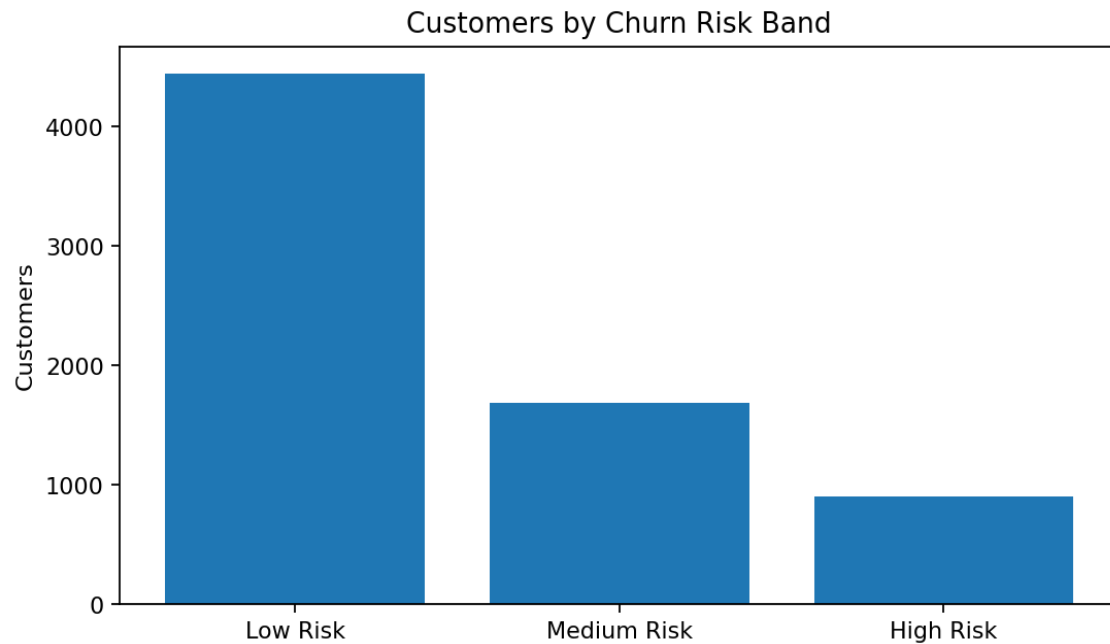
Permutation importance was used so the importance values are comparable at the original business-feature level across model types. Tenure and contract type are the two strongest features in the selected model, followed by internet service, Tech Support, Online Security and Monthly Charges.



| Feature          | Permutation importance |
|------------------|------------------------|
| tenure           | 0.0573                 |
| Contract         | 0.0500                 |
| InternetService  | 0.0118                 |
| TechSupport      | 0.0052                 |
| OnlineSecurity   | 0.0049                 |
| MonthlyCharges   | 0.0038                 |
| PaymentMethod    | 0.0034                 |
| MultipleLines    | 0.0022                 |
| PaperlessBilling | 0.0020                 |
| TotalCharges     | 0.0016                 |

## Risk Scoring & Retention Priorities

The final model scores every customer. The output contains 4,447 low-risk, 1,691 medium-risk and 905 high-risk customers. High-risk customers have an observed churn rate of approximately 75.03% in the scored dataset.



## Recommendations

1. Prioritise high-risk customers for proactive retention outreach.
2. Test incentives that move vulnerable month-to-month customers toward longer contracts.
3. Investigate the experience and friction associated with electronic-check customers.
4. Review fibre-optic pricing, service quality and support because this group shows elevated churn.
5. Strengthen onboarding and early-life engagement for customers in their first 12 months.
6. Promote Tech Support and Online Security bundles where risk is elevated.
7. Combine churn probability with customer value before deciding retention-spend priorities.

## Model Governance Note

The chosen model and probability threshold should be tuned to the economics of the retention campaign. Although Gradient Boosting had the highest ROC-AUC, its churn recall at the default threshold was lower than Logistic Regression and Random Forest. A production deployment should therefore evaluate threshold changes and the relative cost of false negatives versus false positives.

## Power BI Deliverable

The accompanying Power BI source project contains four pages: Executive Overview, Churn Drivers, Customer Segments, and Retention & High-Risk Customers. It is designed to refresh from the project's GitHub-hosted churn\_powerbi.csv file after the GitHub upload is complete.